Question ID: 7a4475df

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question
A function p estimates that there were $2,000$ animals in a population in 1998. Each year from 1998 to 2010, the function estimates that the number of animals in this population increased by 3% of the number of animals in the population the previous year. Which equation defines this function, where $p(x)$ is the estimated number of animals in the population x years after 1998?

Answer
A. $p(x) = 2{,}000(3)^x$

B. $p(x) = 2{,}000(1.97)^x$

C. $p(x) = 2{,}000(1.03)^x$

D. $p(x) = 2{,}000(0.97)^x$

Correct Answer: C

Rationale
Choice C is correct. It's given that a function p estimates that there were $2,000$ animals in a population in 1998 and that each year from 1998 to 2010, the number of animals in this population increased by 3% of the number of animals in the population the previous year. It follows that this situation can be represented by the function $p(x) = a\left(1 + \frac{r}{100}\right)^x$, where $p(x)$ is the estimated number of animals in the population x years after 1998, a is the estimated number of animals in the population in 1998, and each year the estimated number of animals increased by $r\%$. Substituting $2,000$ for a and 3 for r in this function yields $p(x) = 2{,}000\left(1 + \frac{3}{100}\right)^x$, or $p(x) = 2{,}000(1.03)^x$.

Choice A is incorrect. This function represents a population in which each year the number of animals increased by 200%, not 3%, of the number of animals in the population the previous year.

Choice B is incorrect. This function represents a population in which each year the number of animals increased by 97%, not 3%, of the number of animals in the population the previous year.

Choice D is incorrect. This function represents a population in which each year the number of animals decreased, rather than increased, by 3% of the number of animals in the population the previous year.